Independent review of the Topic B proof dependencies

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1 What was reviewed

The review concerns the mathematical meaning of the paired dependency graph, not a new proof of the historical classification inputs and not a measurement of compression. I read the complete current common paper and its four applications, and compared the historical routes with the relevant complete proof bodies of N001, N003, N008 and the fixed revised edition of L030. For N001 the available document is the previously bound authored browser rendering. Its original PDF remains unavailable. The graph records thirty fixed clauses: four, eight, eleven and seven respectively.

The principal organizing proofs in the common paper are correctly separated. The forest intersection argument determines the symmetric subgroups; the laminar partition argument supplies a common representative for a flag. Neither conclusion is substituted for the other. Relative generation uses the successive maximal intersections and the panel choices. The first application retains the earlier classification and uniqueness of inductively minimal pairs. The second retains the stated Whiston inputs and their classification dependence, then supplies the tree and parity calculations. The third proves the necessary finite relative-generation step before using the Frattini ladder, including for an infinite ambient group. The fourth proves its incidence, distance and automorphism assertions, with a separate spectral argument for automorphisms of the uncolored graph.

All arrows were assessed as prerequisites pointing to uses. A displayed route need not be a minimal proof. The parent and branch locators are semantic references, not disjoint quantities that can be added as costs. Attribution, formal verification and diagnostic evidence remain separate from the mathematical arrows. No required enumeration program is hidden in this topic. Retained external costs and unavailable image costs remain unknown, not zero.

2 Five corrections to the graph

The initial frozen graph had no newly discovered manuscript proof defect, but its finer routes needed the following repairs. The revised graph and its exact hashes are bound by the accompanying review record.

1. The propagation branch in N008, Theorem 3.1, uses the maximal chain, the first failed index and earlier containment hypotheses established in the preceding setup. An arrow from that setup was missing. The final contradiction belongs to the assembled theorem; the setup locator is not a claim that the propagation proof depends on its own conclusion.

2. The replacement route for N003, Corollary 2.3, needs the classification of an arbitrary extremal independent family before applying the constructive forest and parity intersection formulas. The full application already contained this classification; the individual output route omitted it.

3. Clause C11 includes containment of the Frattini subgroup in the action kernel for a firm residually weakly primitive geometry. The general identity for the core alone does not establish that containment. The corrected route also includes the Frattini containment theorem. Normality then completes the implication.

4. The retained classification dependence applies to the cited Whiston bound as well as to the omission-subgroup trichotomy. The graph must not make the former appear to use a separately established classification-free proof. This records the retained proof, not logical indispensability of the classification theorem.

5. L030's one-line proof of connectedness of the bipartite double mentions odd cycles but also needs connectedness of the original Kneser graph. The even and odd paths in Lemma 2.2 supply that premise, including the parameter boundary not covered by the strict inequality in the printed Lemma 2.1. The added edge is explicitly an implicit premise, not a fabricated citation.

3 Scope and limitations

The graph preserves the already accepted qualifications. The false printed converse in N003's recall of Whiston is not retained. Its final "immediate" corollary is distinguished from the explicit proof for arbitrary parabolic subgroups in the application. N008's unrestricted infinite generating-family converse is corrected, while the infinite geometric containment theorem is preserved. Its dimension exercise is distinguished from the new written calculation. The L030 corrections concern the exact revised edition and the displayed 63-vertex graph; they do not refute the existence of a different locally Desargues graph.

This is a mathematical and documentary graph review. It does not certify every sentence of the historical papers, reread the original proofs of all retained classifications, or repeat the previously reviewed Lean and finite diagnostics. The structural replay checks bindings, spans, acyclicity and closures. A separate small independent checker protects the five repaired interfaces and recomputes the thirty paired output closures and the four complete application closures. Neither checker replaces the written reading. The acceptance record assigns no text weights, factor, historical priority, new source coverage or new conjecture credit.